Lab Report 06

Course title: Computer Network Laboratory
Course code: CSE-3202
Session: July - December 2026

Date of Submission: *21 October, 2026*

Submitted to-

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Title: Implementation of OSPF Routing

Objective:

- To understand the working principle of OSPF routing.
- To configure OSPF on three routers.
- To establish communication between three different LANs.

Description:

Three routers, three switches and six PCs were added to Cisco Packet Tracer. Two PCs were connected to each switch and each switch was connected to a router. The three routers were connected in a linear topology.

IP addresses, subnet masks and default gateways were configured for all PCs. The router interfaces were also assigned appropriate IP addresses. The network consisted of three LAN networks (192.168.1.0, 192.168.2.0 and 192.168.3.0) and two router-to-router connection networks (10.0.0.0 and 20.0.0.0).

OSPF (Open Shortest Path First) was configured on each router through the CLI using the following commands:

```
router ospf 1
network 192.168.1.0 255.255.255.0 area 0
network 192.168.2.0 255.255.255.0 area 0
network 192.168.3.0 255.255.255.0 area 0
network 10.0.0.0 255.0.0.0 area 0
network 20.0.0.0 255.0.0.0 area 0
exit
```

The appropriate network statements were configured on each router according to its connected networks. OSPF was configured in Area 0. The routers exchanged routing information and learned about remote networks automatically. OSPF uses cost to determine the best route to a destination.

Connection:

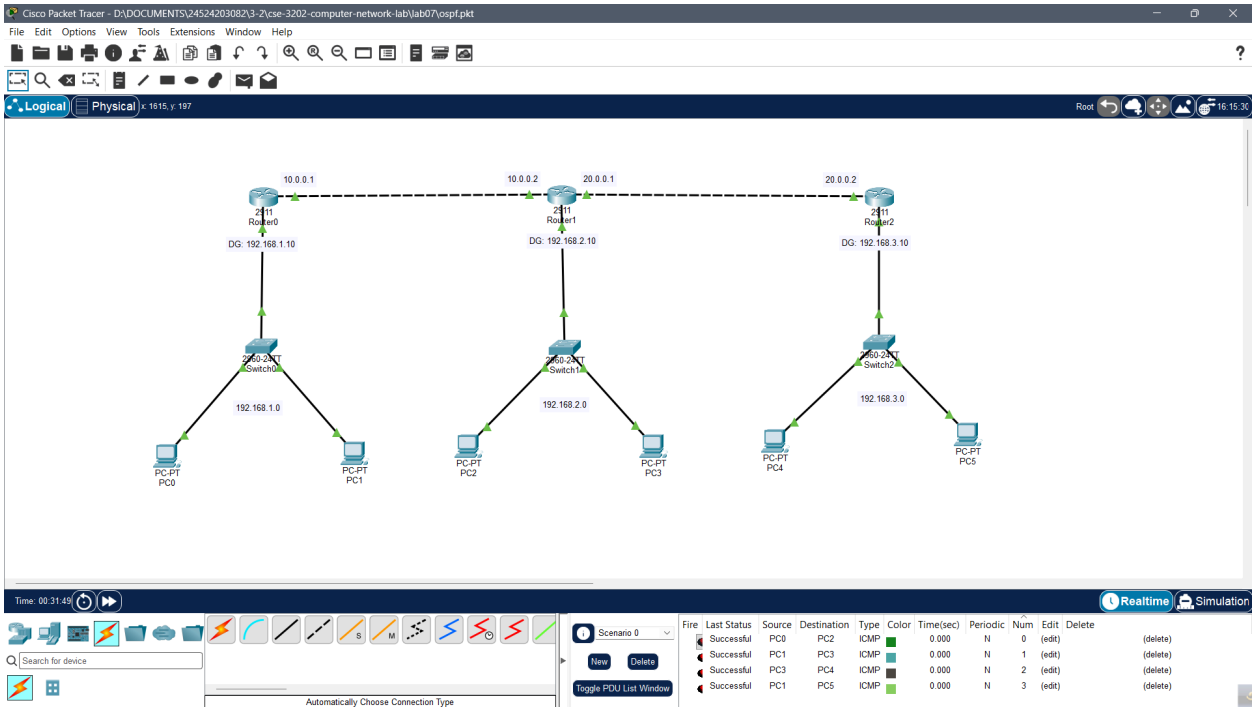


Figure 1: Network Topology for OSPF Routing

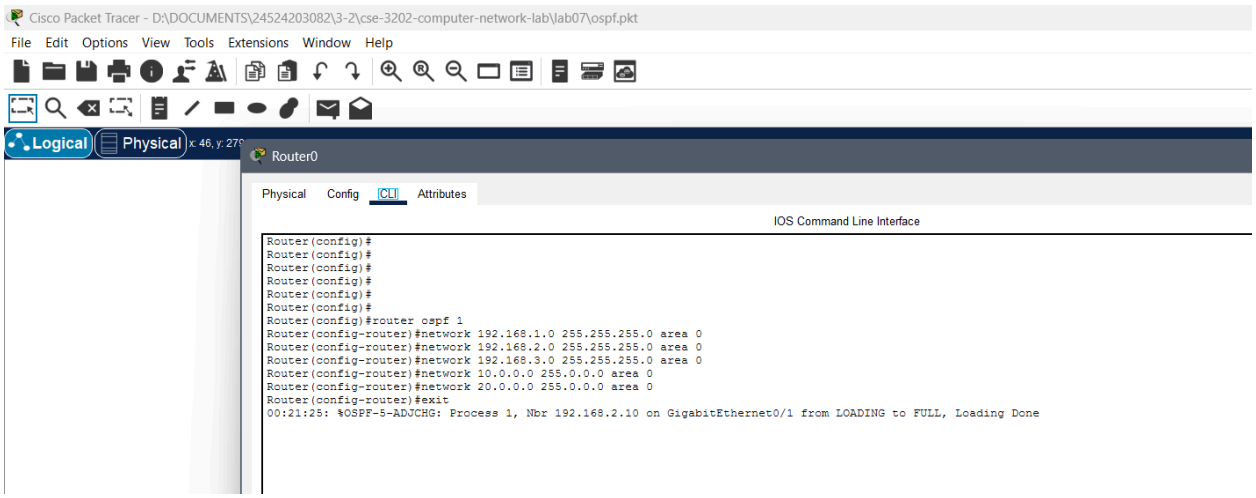


Figure 2: OSPF Configuration on Routers

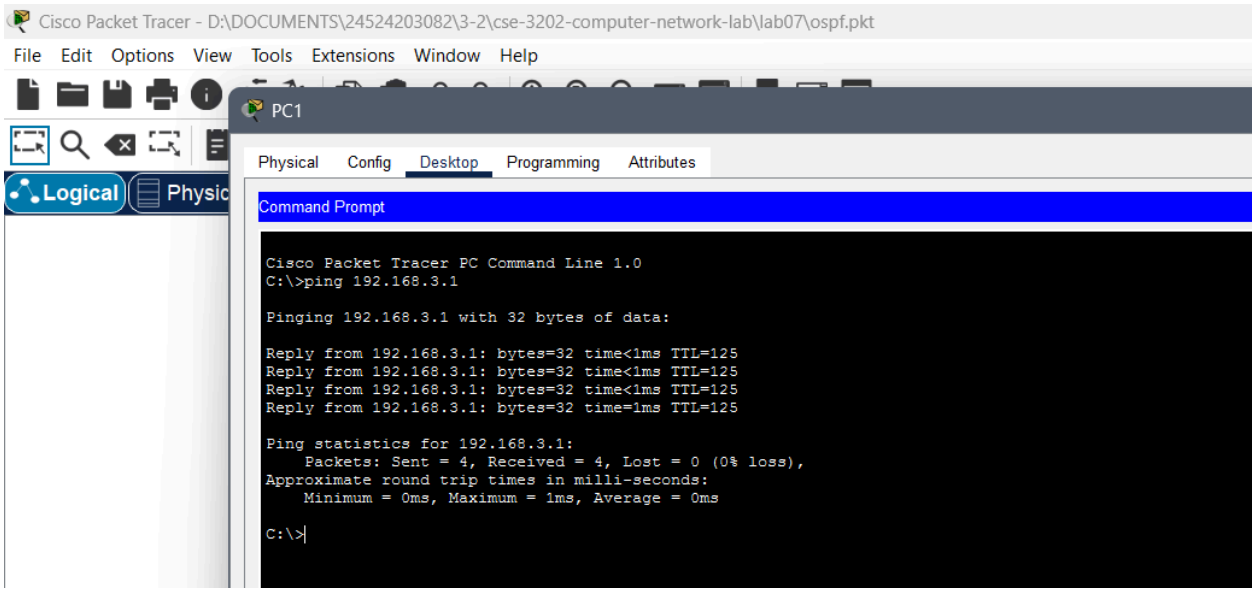


Figure 3: Ping Connectivity Test Between Different LANs

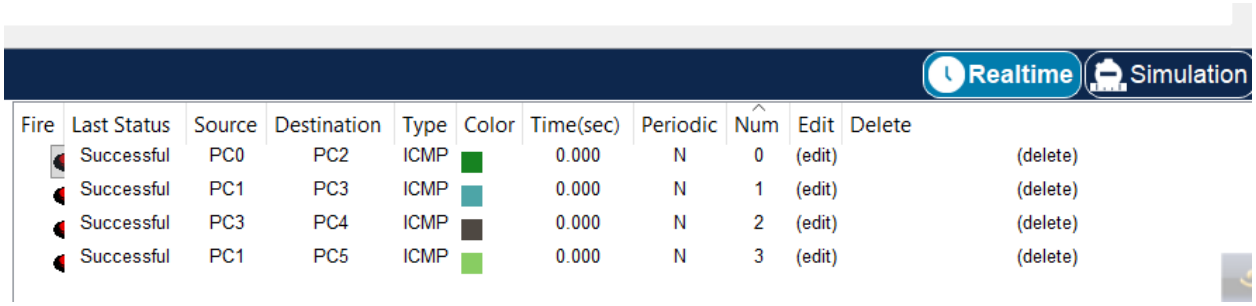


Figure 4: Successful Communication Between LANs